

MS.Net Interview Questions - Simplified

109 What Is: That Every Jobseeker Should Know

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109 .NET Interview Questions

Why are most of the CSE and IT engineers jobless?

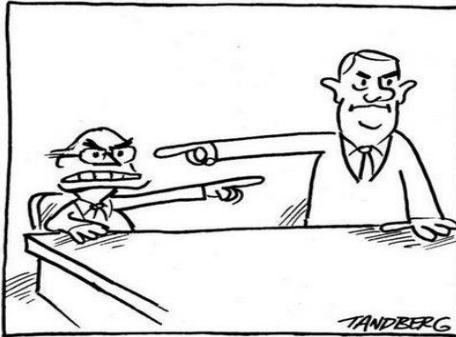


As soon an average 10+2 student gets an admission in any engineering college for IT branch he feels on top of the world and spreads the news to all his relatives that he is going to be a software engineer soon.

He starts enjoying the engineering life. But, slowly he feels that few lecturers are not teaching well and don't have few lecturers at all. Always there will be two subjects as tough on top of their list i.e., Maths and CDS (C and Data Structures).....

Read more at <http://goo.gl/wsRq5o>

Who is responsible and what is the solution for jobless engineers?



When this question is asked to engineers they blame lecturers, HOD and college management. When we turn this question to lecturers, HOD and college management they blame students (that they haven't attended the classes) and the training institutions (that they haven't taught them well in their final sem project). When the same question is asked to training institutions they blame students, lecturers, HOD, college management and they add one more object i.e., IT market and says IT market is down these days due to blaa-blaa in USA.....

Read more at <http://goo.gl/QhxDmd>

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About the Author



ManzoorTheTrainer – MCTS, MCPD and MCT. Planning for Ph.D in the domain of Middleware Technologies with over 14+ years of teaching and over 8+ years of Development Exp. Has an excellent track record as one of the best IT faculty in imparting technology. Makvin's Ace Trainer who believes in giving the real essence of technology to the learner this uniqueness has made me earn a strong foothold in the IT Training Industry.

[Microsoft Certified Professional Profile \(MCP\)](#)

Founder of [ManzoorTheTrainer](#)

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About the Technical Reviewers

I am really thankful to Almighty and all the followers of my video training portal www.ManzoorTheTrainer.com and encouraging me to take a step of writing a technical Interview Questions book on MS.Net i.e., "**109 .NET Interview Questions**".

I am thankful to Mr. Satish Kumar Mandava, my colleague having 6+ yrs of .NET expertise for reviewing this book and giving feedbacks for betterment.

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Chapter 1: MS-SQL Server

Q1. What is Relation in RDBMS?

Ans:

- A table in a database is called as relation.

Q2. What is Tuple?

Ans:

- A Row of a table in database is called as tuple.

Q3. What is RDBMS?

Ans:

- DBMS is a layer of abstraction between application and database.
- Application interact with the Database via DBMS using Structure Query Language(SQL).
- If the database is logically in the form of tables i.e., relation then we call this DBMS as RDBMS.

Q4. What is Primary Key?

Ans:

- Primary Key is a column of a table which is used to identity a row uniquely.
- Primary Key cannot have NULL values.
- Primary Key cannot have duplicate values.

Q5. What is Unique Key ?

Ans:

- Unique Key is a column in a table which does not allow duplicate values like primary key.
- It allows one and only one NULL values.
- A table can have one and only one primary key.
- A table can have one or more unique keys.

Q6. What is Foreign key ?

Ans:

- Foreign key is a column of one table which refers to the Primary Key column of another table.
- It allows NULL values.
- It allows Duplicate values.

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Q7. What is use of Group by clause ?

Ans:

- Group by clause is used to display data by combining or grouping one or more columns.
- Group by clause is always is used in combination with Aggregate Functions.
- Basically Group By clause comes after Where Clause and before Having Clause.
- The typical order of clauses in a query is WGHO (W-Where, G-Group By, H-Having, O-Order By) in most of the queries

Q8. What is Having Clause ?

Ans:

- Having clause is similar to where clause.
- Having clause is used to filter the column generated by aggregate functions.
- As where clause cannot be used with aggregate functions.

Q9. What is the use of DateDiff and DateAdd Functions ?

Ans:

- DateDiff Function is used to find the difference between two dates in terms of years or months or days or hours or minutes or seconds or milliseconds.
- DateAdd Function is used to generate a new date from a specific date by adding or subtracting years or months or days or hours or minutes or seconds or milliseconds.

Q10. What is Join ?

Ans:

- Join is a Clause used to extract the data from two or more tables based on a specific columns or Primary - Foreign Key relationship
- There are various Joins like Inner Join, Left Outer Join, Right Outer Join, Full Outer Join, Self Join.

Q11. What is Inner Join ?

Ans:

- Inner join is a join to get data based on matched rows from left table and matched rows from right table.

Q12. What is Left Outer Join ?

Ans:

- Left Outer join is a join to get all the rows from left table and matched rows from right table.

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Q13. What is Right Outer Join ?

Ans:

- Right Outer join is a join to get matched rows from left table and all rows from right table.

Q14. What is Full Outer Join ?

Ans:

- Full Outer join is Join to get all rows from left table and all rows from right table.

Q15. What is Self Join ?

Ans:

- Self join is Join which is applied on same table.
- Joining a table with itself with different table aliasing.

Q16. What is Stored Procedure ?

Ans:

- Stored Procedure is precompiled set of one or more SQL statements which performs some specific task.
- Store Procedure should be executed stand alone using EXEC.
- Store Procedure can return multiple parameters.
- Store Procedure can be used to implement transaction.

Q17. What is Functions ?

Ans:

- A Functions is a section of code that performs an independent task.
- Function cannot be executed stand alone, it is always used in a query like with select clause.
- Functions can return one and only one parameter.
- We cannot implement transaction using functions.

Q18. What is Transaction?

Ans:

- Transaction is a set of operation (SQL Queries) which must be executed as a group.
- If at all a single operation is failed then all the previous operation of that group must be rolled back i.e., undone.

Q19. What is Scope_Identity() function?

Ans:

- Scope_Identity function returns the recent generated identity value (that is auto incremented values) for that particular scope i.e., user.

Chapter 2: MS.Net Framework

Q20. What is dotnet framework?

Ans.

- Dotnet framework is a language neutral platform which is used to develop any kind of application like web application, windows application, mobile application, web services, etc.,
- It is called as language neutral because multiple programming languages uses the same class library which is called as frame class library or base class library.
- Latest version of .Net as on DEC 2013 is 4.5

Q21. What is CLR ?

Ans.

- CLR stands for common language runtime and it provides a runtime environment for dotnet applications.
- It supports memory management, exception management, type safety, code management, MSIL to native code conversion.

Q22. What is MSIL or assembly or component ?

Ans.

- MSIL stands for Microsoft Intermediate Language.
- The compiled version of c# code is called as assembly or MSIL or IL or component.

Q23. What is the difference between exe and dll ?

Ans.

- .exe and dll are the compiled version of c# code which are also called as assemblies.
- .exe is a stand alone executable file, which means it can executed directly.
- .dll is a reusable component which cannot be executed directly and it requires other programs to execute it.

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Q24. What is JIT compiler?

Ans.

- JIT stands for just in time compiler.
- It is major component of CLR which is responsible for converting MSIL (microsoft intermediate language) to native code.
- Native code is basically is a binary code which is understood by windows operating system.

Q25. what is CLS ?

Ans.

- CLS stands for common language specification.
- It is a set of rules followed by a programming language which allow them to use dotnet framework class library or base class library or class library.

Q26.what is CTS ?

Ans.

- CTS stands for common type system.
- These are standard data types defined for all the programming language supported by dotnet framework It is subset of CLS.

Q27.What is GC ?

Ans.

- GC stands for garbage collection.
- It is the program or part of a program called thread that runs in background of the application to free the memory occupied by the unused objects.
- It is also a major component of CLR.

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Chapter 3: C#.Net

Q28. What is object ?

Ans:

- Any real time entity is called as an object.
- Every object consist of state(look and feel) and behaviour(what is does).
- States are called as fields and behaviors are called as methods.

Q29. What is a class ?

Ans:

- Class is a blueprint of an object.
- It consist of fields which are not allowed to access from outside the class.
- It also consist of methods which are allowed to access from outside the class.

Q30. What are access modifiers ?

Ans.

- Access modifiers are the modifiers given to fields or methods to implement access restrictions.
- If a member of a class (fields or methods) is declared as public it can be accessed from outside of the class.
- If a member of a class(fields or methods) is declared as private it can not be accessed from outside the class.

Q31. What is the difference between value type and reference type ?

Ans.

- Value type are the data type variables for which the memory is allocated at compile time.
- This concept is also called as early binding or static binding.
- Reference type are the data type variable for which the memory is allocated at runtime.
- This concept is also called as late binding or dynamic binding

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Q32. What is constructor?

Ans.

- Constructor is a special method which has same name as class name.
- It does not have return type, but can have parameters.
- It should be public; in some situations it can be private.
- Default Constructor of a class is without any parameters.
- Constructor is invoked automatically when we create an object.
- You cannot invoke constructor explicitly.

Q34. What is the use of constructor?

Ans:

- Constructor is used to initialize the object.

Q35: What is constructor overloading?

Ans:

- If we have more than one constructor in a class with different parameters then we called it as constructor overloading.

Q36: What is Method overloading?

Ans:

- If in a class there is more than one method, with the same name but different parameters then we call this as method overloading.

Q37. What is compile time or static polymorphism?

Ans:

- In method Overloading the code for the method is generated at compile time itself. Hence it is called as static polymorphism.

Q38. What is *this* keyword?

Ans:

- this keyword is a default object of current class.
- In what ever class we use "this" keyword it acts like its object

Q39. What is the use of *this* keyword?

Ans:

- 'this' keyword has two uses.
- 'this' keyword is used to refer the members of same class.
- It is also used to invoke a constructor from another constructor of the same class.

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Q40. What is static variable?

Ans:

- Static variable is a variable which is declared by static keyword.
- Static variable is global for all the objects of that class in which it is declared.
- Memory for static variable is allocated at compile time and it is allocated only once.
- All the objects of that class can access that variable or same memory location.
- Static variable cannot be access with the object we can access static variable directly with the class name.

Q41. What is static constructor?

Ans:

- Static constructor is a constructor which is declared with the keyword as static.
- Static constructor is use to initialize static variables (private).
- Static constructor does not have any access modifiers (i.e,Public,Private).
- Static constructor is parameter less.
- There can be one and only one static constructor in a class.
- Static Constructor is invoked before creation of any object.

Q42. What is static method?

Ans:

- Static method is a method which is declared with the keyword static.
- There can be n number of static methods in a class.
- Static methods can access only static variables.
- Static method can access by Class name.

Q43. What is static class?

Ans:

- Static class is a class which is declared with static keyword.
- Static class should contain only static methods.
- Static class does not allow creation of class reference or object

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Q44. What is a property?

Ans:

- Properties are special kind of methods to access any field in a secured way.
- It contains two kind of methods i.e., getter and setter
- Getter method is to read the value of the private field
- Setter method is to write the value of the private field
- It looks like we are accessing the fields directly. But, internally it works as methods where we can perform some kind of validations

Q45. What is Namespace?

Ans:

- Namespaces are logical group of classes and namespaces.
- It is used for avoiding name clashes between two sets of code.

Q46. What is inheritance?

Ans:

- Deriving a new class from an existing class is called Inheritance
- The class from which other classes derive fundamental functionality is called a base class.
- The class which derives functionality from a base class is called a derived class.

Q47. What is single inheritance?

Ans:

- If a derived class has a single base class then it is called as single inheritance
- i.e., $A(\text{Base}) \rightarrow B(\text{Derived})$

Q48. What is multi level inheritance?

Ans:

- If a class is derived from a class which is already a derived class then it is called as multi level inheritance.
- i.e., $A(\text{Base}) \rightarrow B(\text{Derived}) \rightarrow C(\text{Derived})$

Q49. What is multiple inheritance?

Ans:

- If a class is derived from more than one class then it is called as multiple inheritance.
- i.e., $A(\text{Base}) \rightarrow B(\text{Derived}) \rightarrow C(\text{Derived})$
- C# does not support multiple inheritance using classes.

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Q50: What is base keyword?

Ans:

- base keyword is a default object of its base class.
- In what ever class we use "base" keyword it acts like an object of its base class.

Q51: What is the use of base Keyword?

Ans:

- base keyword has two uses
- base keyword is used to invoke a constructor of base class from the constructor of derived class. It is also called as constructor chaining.
- It is also used to refer the members of base class from derived class.

Q52: What is method overriding?

Ans:

- If a method in the derived class has the same signature as that of in the base class with different implementation then we say that method in the derived class overrides the method of base class. This process is called as Method Overriding.
- The method of the base class should be declared as virtual and method in derived class should be declared as override.

Q53: What is a virtual method?

Ans:

- A virtual method is the method which is declared with the keyword virtual and it can be overridden by the derived class method by using override keyword.
- Its upto the derived class weather to override it or not.

Q54: What is a sealed method?

Ans:

- A sealed method is a method which is declared with the keyword sealed and it cannot be overridden by the derived class.
- But we can invoke this method.
- Sealed keyword is always used in combination with override keyword.

Q55: What is a sealed class?

Ans:

- A sealed class is the class which is declared with the keyword sealed and it cannot be derived by other classes.
- But we can create the objects of sealed class.

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Q56: What is method hiding?

Ans:

- If a method in the derived class has the same signature as that of in the base class with different implementation then we say that method in the derived class hides the method of base class. This process is called as Method Hiding.
- The method of the base class should not be declared as virtual and method in derived class should be declared as new.
- We cannot implement runtime polymorphism using method Hiding.

Q57: What is abstract method?

Ans:

- An abstract method is a method which is declared with the keyword abstract.
- It does not have body.
- It should be implemented by the derived class.
- If a method is abstract then the class should be abstract.

Q58: What is abstract class?

Ans:

- It is the blue print of the problem.
- It says what to do but not how to do.
- An abstract class is the class which is declared with the keyword abstract.
- Basically it contains at least one abstract method.
- It's object cannot be created.
- But it's reference can be created.
- It is used to implement runtime polymorphism.

Q59: What is runtime or dynamic polymorphism?

Ans:

- Deriving the abstract class by one or more classes and overriding all the abstract methods.
- Creating the reference of abstract class and making it an object of derived class based on user's input.
- And calling the method of abstract class.
- The method's code gets generated from derived class at runtime based on user's input.
- This complete process is called as runtime polymorphisms.

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Q60: What is interface?

Ans:

- An interface is a pure abstract class.
- It is a reference type and it contains only abstract members.
- Interface's members can be
 - Events
 - Methods
 - Properties
 - Indexers.
- Any implementation must be placed in class that implements them.
- The interface can't contain
 - Constants
 - Data fields
 - Constructors
 - Destructors
 - Static members.
- All the member declarations inside interface are implicitly public and abstract.
- We can implement multiple inheritance using interfaces.

Q61: What is a namespace?

Ans:

- Namespaces are logical segregation of classes and namespaces.
- Namespaces allows you to organize your classes.
- They also provide assistance in avoiding name clashes between two sets of code.

Q62: What is a partial class?

Ans:

- Partial class is a class which is declared with partial keyword.
- Its implementation can be spread over multiple files i.e., multiple classes prefixed with partial keyword with same name and different methods in different files, acts as a single class with all the methods.

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Q63: What is a collections?

Ans:

- To store multiple **fixed** number of elements or objects we use arrays.
- Collection is an alternate way of storing multiple **variable** number of elements or objects.
- Like Stack, Queue, List , etc., are collection classes present in System.Collection namespace
- There is no type safety in collection classes i.e., in a collection object of a Queue we can store few integers and few strings.
- We need to type cast the elements while accessing it.

Q64: What is generic collection class?

Ans:

- Generic collection is an alternate way of storing multiple **variable** number of elements or objects.
- Like Stack, Queue, List , etc., are generic collection classes present in System.Collection.Generic namespace
- There is type safety in generic collection classes i.e., in a generic collection object of a Queue we cannot store few integers and few strings. Hence it is strongly typed.
- We need not to type cast the elements while accessing it.

Q65: What is exception and how do we handle it?

Ans:

- Runtime error is called as an exception.
- Exception handling is an in built mechanism in .NET framework to detect and handle runtime errors.
- C# provides three keywords try, catch and finally to do exception handling.
- The try encloses the statements that might throw an exception.
- The catch handles an exception if one exists.
- The finally can be used for doing any clean up process.

Q66: What is delegate?

Ans:

- A reference to a method is called as delegate.

Q67: What is event?

Ans:

- Any action performed is called as event.
- Events are handled with the help of delegates.

Chapter 4: ASP.Net

Q68: What is Asp.Net page lifecycle?

Ans:

- When we request an ASP.NET page, the page goes through a life cycle in which it performs a series of processing steps.
 - **Page request** → Checks whether a cached version of the page can be sent in response without running the page.
 - **Start** → Sets the page properties such as Request, Response, IsPostBack and UICulture.
 - **Initialization** → Each control's UniqueID property is set. A master page and themes are also applied to the page if applicable.
 - **Load** → During load, if the current request is a postback, control properties are loaded with information recovered from viewstate and control state.
 - **PostBack event handling** → If the request is a postback, control event handlers are called.
 - **Rendering** → Before rendering, view state is saved for the page and all controls. During the rendering stage, the page calls the Render method for each control.
 - **Unload** → The Unload event is raised after the page has been fully rendered and page properties such as Response and Request are unloaded and cleanup is performed.
- We have various events that a page undergoes during the above processing steps like PreInit, Init, InitComplete, PreLoad, Load, etc.,

Q69: What is IsPostBack property?

Ans:

- It is a property which tells us whether the page is being loaded for the first time or is being loaded in due to a postback.
- i.e., if IsPostBack is false then it is being loaded for the first time else it is being loaded due to a postback

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Q70: What is server control?

Ans:

- Any control with *runat* attribute set to "server" is a server control.
- Basically it is the job of asp.net web server to convert server control into its equivalent html control.

Q71: What is a validation control?

Ans:

- Any control which is used to validate the fields of the form is called as a validation control.
- i.e., RequiredFieldValidator, RangeValidator, etc.,

Q72: What is the validation control used to validate date?

Ans:

- Compare Validator by setting *type* property to "Date".
- We can also achieve this at client side by setting the TextMode of a textbox to "Date" which is an HTML5 feature.

Q73: What is authentication?

Ans:

- Checking the genuinity of a user is called as authentication.
- Like asking username and password

Q74: What is windows authentication?

Ans:

- Checking the genuinity of a user using the login credentials of windows OS is called as windows authentication.
- i.e., there will no separate window for username and password.
- By default our web apps are set to windows authentication.

Q75: What is forms authentication?

Ans:

- Checking the genuinity of a user using separate login credentials is called as forms authentication.
- i.e., there will be a separate form for username and password.
- We can set forms authentication by setting the mode attribute of Authentication tag in web.config
- i.e.,`<system.web> <authentication mode="Forms" /> </system.web>`

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Q76: What is authorization?

Ans:

- Giving access rights to a user after authentication is called Authorization.
- It is basically achieved with the help of roles.

Q77: What is gridview and can we perform insert operation in a gridview?

Ans:

- It is visual representation of a table on web page where each column represents a field and each row represents a record.
- The GridView control enables you to select, sort, delete and edit these items.
- It does not provide Insert operation.

Q78: What is the process or the event used to display sum of a column in a grid footer?

- We can perform the sum while gridview is loading by writing the code in RowDataBound event.
- GridView.RowDataBound event occurs when every row is bound to gridview.

Q79: What is the process of adding dropdown list in a gridview column?

- Gridview allows us to add template columns and we can use that to add dropdown list, textbox, radio button list or any server control as column.
- Or we can also convert an existing column into a template column and change it to drop down list.

Q80: What is a user control?

- Some time you may need functionality in a control that is not provided by the built-in ASP.NET Web server controls.
- In this case you can create your own controls called as User controls.
- User controls are combination of various Web server controls and some code implemented for it.
- You can then treat the user control as new control and use on any page by dragging and dropping it from project folder.

Q81: What is a master page?

- A single page which contains the common content of all the pages or group of pages.
- Like header, menu and footer is common on all the pages and can go into master page.

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Q82: What is state management?

Ans:

- HTTP is a stateless protocol i.e., Server will not maintain the state of the client.
- In simple terms we cannot carry any value or data from one post back to another post back on same page or different pages by default.
- To achieve this we follow some techniques called as state management techniques.

Q83: What is client side state management?

Ans:

- To carry any value or data from one post back to another post back on same page or different pages we store that data on client machine or computer.
- To achieve this we follow some techniques called as client side state management techniques.
- Following are client side state management techniques Cookies, QueryStrings (URL), Hidden fields and View State.

Q84: What is a cookie variable?

Ans:

- A cookie is a small piece of text stored on user's computer. Usually, information is stored as name-value pairs.
- Can be used to carry the value throughout the application.
- If you want to store information use the below code
 - `Response.Cookies["UserId"].Value=username;`
- If you want to retrieve information use the below code
 - `string username = Request.Cookies["UserId"].Value;`

Q85: What advantages and disadvantages does a cookie variable has?

Ans:

- Advantages:
 - Simplicity
- Disadvantages:
 - Cookies can be disabled on user browsers.
 - Cookies are transmitted for each HTTP request/response causing overhead on bandwidth
 - Inappropriate for sensitive data

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Q86: What is query string?

Ans:

- A query string is the part of a URL that contains data to be passed from one page to another page.
- Can be used to carry the value from one page to another page.
- They are passed along with URL in clear text which can be encrypted.
- If you want to pass data using query string use the below code
 - `Response.Redirect("productdetails.aspx?productid=4")`
- If you want to retrieve data from query string use the below code
 - `productid=Request.Params["productid"];`

Q87: What advantages and disadvantages does a Query string has?

Ans:

- Advantages:
 - Simple to Implement
- Disadvantages:
 - Human Readable
 - Client browser limit on URL length
 - Easily modified by end user

Q88: What is hidden field?

Ans:

- Hidden field is just like a standard control for which you can set its properties.
- Whenever a page is submitted to server, hidden fields values are also posted to server along with other controls on the page.
- Can be used to carry the value on the same page between different postbacks.
- As its name says, these fields are not visible on the page.

Q89: What advantages and disadvantages does a hidden field has?

Ans:

- Advantages:
 - Simple to implement for a page specific data
 - Can store small amount of data so they take less size.
- Disadvantages:
 - Inappropriate for sensitive data
 - Hidden field values can be intercepted (clearly visible) when passed over a network

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Q90: What is viewstate variable?

Ans:

- View State is the default feature of Asp.net server controls.
- Can be used to carry the value or data on the same page between different postbacks.
- If you want to store data in ViewState use below code
 - `ViewState["myviewstate"] = myValue;`
- If you want to read data from ViewState use below code
 - `Response.Write(ViewState["myviewstate"]);`

Q91: What advantages and disadvantages does a viewstate variable has?

Ans:

Advantages:

- Simple for page level data
- Encrypted
- Can be set at the control level

Disadvantages: –

- Overhead in encoding ViewState values
- Makes a page heavy

Q92: What is server side state management?

Ans:

- To carry any value or data from one post back to another post back on same page or different pages we store that data on server machine or computer.
- To achieve this we follow some techniques called as server side state management techniques.
- Following are server side state management techniques Session Variables, Application Variables.

Q93: What is session state variable?

Ans:

- ASP.NET session state enables you to store and retrieve values for a user.
- Can be used to carry the value throughout the application.
- If you want to store data in Session State use below code
 - `Session["mydata"] = myValue;`
- If you want to read data from ViewState use below code
 - `Response.Write(Session["mydata"]);`

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Q94: What advantages and disadvantages does a session variable has?

Ans:

- Advantages:
 - Fastest mode
 - Simple configuration
- Disadvantages:
 - Session data will be lost if the worker process or application domain recycles.

Q95: What is Application state variable or Application object?

Ans

- Application state variable is used to store data which is visible across entire application and shared across multiple users.
- Data which needs to be persisted for entire life of application should be stored in application object.
- It's a great place to store data which does not changes frequently.
- We should write to application variable only in Application_Start event.
- If you want to store data in Application variable use below code
 - `Application.Lock();`
 - `Application["mydata"]="mydata";`
 - `Application.Unlock();`
- If you want to read data from Application variable use below code
 - `Response.Write(Application["mydata"]);`

Q96: What advantages and disadvantages does an application variable has?

Ans:

- Advantages of using application state are:
 - Simple implementation.
 - A single copy of the information for all users to access.
- Disadvantages of using application state are:
 - Application state will be lost if the Web server process containing it is crash, upgrade, or shutdown.
 - Application state requires server memory, which can affect the performance of the server.

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Q97: What is global application class?

Ans:

- It is a class which contains global events of the application like Application_Start, Application_End, Application_Error, Session_Start, Session_End, etc.,
- We can perform some specific tasks in those events.
- We can have one and only one global application class in a web application.

Q98: What is web.config file?

Ans:

- It is an xml file which contains the configuration settings of the web application like target framework version, connection string, authentication type, providers, smtp settings, etc.,
- We can have multiple web.config files in a web application i.e., one for each folder to give access rights to that folder.
- We need not compile the complete project if we want the changes of web.config file to reflect in the application.

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Chapter 5: ADO.Net

Q99: What is Ado.Net?

Ans:

- ADO stands for Active Data Objects.
- It is a data access component for .Net application.
- i.e., It contains a set of class that can be used to access data from the database like SqlConnection, SqlCommand, etc.,

Q100: What is connection string?

Ans:

- A connection string is a string used to connect to the database.
- The major parts of connection string are
 - Data Source (i.e.,Sql Server Name)
 - Initial Catalog (i.e.,Database Name)
 - Security
 - Integrated (Uses windows authentication)
 - Sql Server (Separate User ID and Password)

Q101: What is ADO.Net data provider?

Ans:

- The ADO.NET Data Provider model provides an interface in the .NET Framework for accessing data from various data bases.
 - For SQL Server provider is System.Data.SqlClient (Inbuilt)
 - For Oracle provider is System.Data.OracleClient (Third Party)
 - For MySql provider is System.Data.MySqlClient (Third Party) , etc.,

Q102: What is connected architectural data access?

Ans:

- The architecture in which as long as we perform the operations on the database we keep the connection open and once operations are done then the connection is close. Such architecture is called as connected architecture.
- SqlDataReader works on connected architecture.

Q103: What is SqlDataReader?

Ans:

- SqlDataReader is a class which works on connected architecture.
- It is used to read the data from database after executing the command.

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- It reads one row after the other in the form of stream of bytes (0s and 1s)
- It is read only (You cannot edit the record which is in datareader)
- It is forward only (You cannot read the data once it is used)

Q104: What is disconnected architectural data access?

Ans:

- The architecture in which we open the connection initially then load the data in an object(Dataset) and close the connection. (Achieved with the help of SqlDataAdapter).
- Now we can perform the CRUD operations on the object which is not connected to database.
- Finally we update the object to the database after performing operations.(Achieved with the help of SqlDataAdapter)
- Such architecture is called as disconnected architecture.
- Dataset works on connected architecture.

Q105: What is DataSet?

Ans:

- Dataset is collection on datatables and xml files.
- Tables and xml files in dataset are stored in the form of an array.
- We can perform insert, update, Delete and select operations on dataset.
- Dataset works on disconnected architecture.

Q106: What is SqlDataAdapter?

Ans:

- SqlDataAdapter is a class which is responsible for opening the connection and loading the data in an object(Dataset) and close the connection (i.e.,Filling the dataset).
- It is also responsible for updating the dataset to the database after performing operations.
- SqlDataAdapter falls under connected architecture and used for implementation disconnected architecture.

Q107: What is untyped DataSet?

Ans:

- When dataset is filled with datatable only data and table structured is copied but constraints are not applied on it. i.e., unique key, primary key, foreign key, etc.,
- If you want you can create the constraints manually.
- Such a dataset is called as untyped dataset.

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Q108: What is typed DataSet?

Ans:

- A dataset which has all the constraints(i.e., unique key, primary key, foreign key, etc.,) defined on it with the help of xml file i.e., xsd (xml schema definition) is called as typed dataset.
- In other words a dataset with xsd file is called as typed dataset.

Q109: What is transaction and how do we implement transaction in ADO.NET?

- Transaction is a set of operation(SQL Queries) which must be executed as a group.
- If at all a single operation is failed then all the previous operation of that group must be rolled back i.e., undone.
- We can implement transactions in ADO.NET using SqlTransaction class by calling BeginTransaction on the SqlConnection object.
- Committing or aborting the transactions are performed on the SqlTransaction object.

Thanks And All The Best For Your Interviews 😊